Deterministic-First AI-Assisted Assessment for the Recognition of Prior Learning: An Offline-First, Contract-Driven Platform for India's Informal Workforce

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Abstract—Recognition of Prior Learning (RPL) is the principal instrument for certifying the estimated 85% of India's informal-sector workers who acquired trade skills on the job but hold no formal credentials; yet only 4.69% of the workforce has received formal skill training. RPL assessment today is largely manual, subjective, and regional-language-blind: assessors grade practical work against National Occupational Standards (NOS) by eye, leading to reported inter-assessor variations of 15–22%, while workers in low-connectivity regions are effectively excluded from remote assessment. This paper presents a deterministic-first, contract-driven backend platform for AI-assisted RPL assessment. Its contributions are: (i) a shared, role-guarded assessment state machine that mirrors the official NSQF Advise→Assess→Audit flow; (ii) a deterministic-first analysis engine that scores regional-language narration and evidence against bilingual keyword rubrics derived from official Qualification Packs, with an explicit confidence router that escalates to LLM/vision/ASR adapters or to human assessors; (iii) an offline-first synchronization contract (client UUIDs, Idempotency-Key, append-only evidence, last-writer-wins versioning) with chunked resumable uploads; (iv) a SHA-256-locked append-only evidence chain with a public certificate verifier page; and (v) four NSQF-aligned trade packs totalling 32 NOS and 569 performance criteria. The implementation is fully contract-driven (37 OpenAPI paths, 141 schemas) and validated by 124 automated tests.

Index Terms—Recognition of prior learning, NSQF, computer-assisted assessment, offline-first architecture, digital evidence chain, multilingual NLP.

I. Introduction

India's workforce is overwhelmingly informal: an estimated 80% of workers are employed in the unorganized sector, and roughly 85% of these acquired their skills through informal and non-formal channels rather than structured training [1], [2]. Formal vocational training reaches only 4.69% of the workforce, against 68% in the United Kingdom, 75% in Germany and 96% in the Republic of Korea [1]. Recent analysis places informal employment at 91% of the workforce [3]. These workers possess real, marketable competence but no credentials that unlock formal employment, migration, or wage progression.

Recognition of Prior Learning (RPL), defined in the National Skills Qualification Framework (NSQF) notification of 27 December 2013 as "the process of recognizing previous learning, often experiential, towards gaining a qualification" [4], is the designated instrument for certifying such competence. The national RPL policy prescribes three stages—Advise, Assess and Audit—and explicitly recommends technology-based assessment "so that costs and subjectivity can be minimized", including assessment "in regional and local languages" [1], [5]. Nevertheless, deployed RPL assessment remains manual: candidates travel to assessment centres, perform scripted tasks under an observer, and their performance is graded by eye against the performance criteria (PCs) of the relevant NOS and Qualification Pack (QP).

Three structural gaps motivate this work. (G1) Subjectivity and throughput. Practical scoring is dominated by holistically scored tasks with reported inter-assessor variation of 15–22%, and an assessor handles roughly 20–25 candidates per day, making national scale (tens of lakhs of RPL candidates under schemes such as PMKVY 4.0, which trained 27.08 lakh candidates across 38 sectors by December 2025 [6]) prohibitively expensive. (G2) Language exclusion. Hindi and Marathi speaking workers—and speakers of the many other Indian languages—cannot narrate or be assessed in their own language. (G3) Connectivity. Large video evidence cannot be uploaded reliably from rural networks, and there is no legally meaningful evidence trail that a verifier can check later.

We address these gaps with a backend platform whose central design position is deterministic-first: the awarding decision never depends on an unverifiable black box. Machine-verifiable rule-based and keyword-driven scoring computed from the official QP rubric happens first; learned components (LLM paraphrase, vision, ASR) are optional, explicitly routed, and always auditable. Contri-

butions are:

1) a shared, contract-driven assessment state machine (12 states) that encodes NSQF RPL workflow with role-guarded transitions;
2) a deterministic-first, bilingual analysis engine with explicit confidence routing to optional AI adapters or to human assessors;
3) an offline-first synchronization contract and resumable evidence uploads;
4) a SHA-256 evidence chain with certificate QR verification by the public;
5) four NSQF-aligned trade packs (Electrician, Welder, Plumber, Mason) with bilingual keyword rubrics, validated as machine-readable contracts;
6) an evaluation harness of 125 automated tests over an OpenAPI surface of 37 paths and 141 schemas.

II. Background and Related Work

A. RPL in India and the NSQF

The NSQF organizes qualifications into ten levels characterised by five descriptor domains (process, professional knowledge, professional skill, core skill, and responsibility). Qualification Packs (QPs)—several thousand of which are published by the industry-led Sector Skill Councils into the National Qualification Register—decompose each job role into 3–7 NOS, and each NOS into 5–30 assessed thinkable performance criteria (PCs) [7], [8]. International practice in recognising non-formal and informal learning is documented extensively [10], [11]. Two features of the policy landscape directly shape our design. First, UNESCO's country case study recommends bridge training and part qualification for candidates who demonstrate most but not all competences [1]. Second, the NSQF notification text and NCVET RPL guidelines specify a minimum 70% per-NOS pass standard, with practical components typically weighted at 70–80% of the total [5]. Our engine therefore computes per-NOS scores, applies the 70% threshold per NOS, and emits a deterministic bridge-training recommendation when a candidate falls short on a minority of PCs—an automated counterpart of the recommended "bridge module".

B. Existing platforms and their gaps

The Skill India Digital Hub (SIDH) provides central registration and course discovery, and PMKVY 4.0 drives macro RPL tracks [9], [6], but both rely on manual profile entry, human pre-screening and paper-style assessment records; none offers regional-language narration, objective rubric scoring computed from the QP itself, offline evidence capture, or a public verifier for the issued credential. The National Qualification Register is a registry of qualifications, not an assessment engine [7]. Our platform complements these by computing the assessment from the same QP files that the councils publish.

C. AI-assisted assessment

Four literature families inform our AI layer. (i) Item Response Theory underlies adaptive questioning but requires item banks calibrated on large populations—unavailable for RPL—so we use deterministic adaptive routing with fixed, contract-defined rubrics instead [12]. (ii) Multilingual text embeddings such as LaBSE enable cross-lingual similarity for text-to-QP matching in regional languages [13]. (iii) Speech recognition, e.g. Whisper, transcribes regional-language narration [14]. (iv) Object detection, e.g. YOLO, can check safety items in candidate video evidence [15]. Reported benchmarks (94.2% mean average precision for PPE detection; 89.7% dialect-to-standard-code alignment) are dataset-dependent; our architecture therefore treats every learned component as a non-authoritative adapter whose output is advisory to the deterministic score and the human assessor.

D. Offline-first synchronization

Progressive Web App (PWA) offline capabilities—service workers and local storage—are standard [16]. For multi-device evidence capture, convergent replicated data types (CRDTs) provide a principled conflict model [17]; in a worker–assessor workflow the practical compromise is last-writer-wins (LWW) with monotonic versions plus idempotent, UUID-keyed writes [18]. Resumable chunked upload is standardised by the tus protocol [20]. We adopt all three, and further require that evidence records be append-only and hash-chained, aligning with India's Bharatiya Sakshya Adhiniyam (BSA) 2023, whose Section 63 makes cryptographic hash-locking a precondition for the admissibility of electronic records [21].

III. System Architecture

The platform is a backend service with a contract-first design: the OpenAPI specification and its Zod schemas are the single source of truth—the schema is the contract, and every endpoint and document type is validated against it at runtime (37 paths, 141 schemas, generated from 9 contract modules) [19]. Figure 1 shows the component layout: a PWA client (the offline-first capture surface) talks over HTTPS to an Express 5 API tier; an orchestration service drives the state machine; an analysis service implements the deterministic-first engine; queue workers (BullMQ on Redis, with an in-process inline driver for single-host deployments) execute audits, report rendering and certificate issuance asynchronously; MongoDB (Mongoose 9) persists documents; and PDFKit plus a QR encoder emit the certificate with an embedded verification link. A media gateway implements the tus protocol for chunked, resumable uploads of video and photo evidence [20]. All inter-component calls are typed end-to-end; role-based access control (worker, assessor, authorized assessor, admin, and public roles) guards every transition and every read.

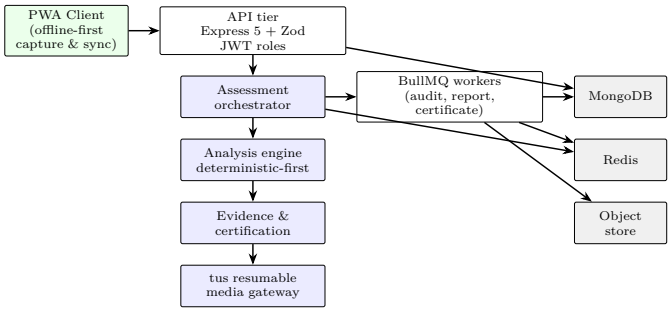


Fig. 1. Component architecture. All workflows are driven through the API tier; every service is typed against the OpenAPI/Zod contract.

A. The synchronization contract

The backend exposes a small set of invariants that the client implements for offline-first operation, enforced server-side:

- Client UUIDs. Every assessment, evidence item, and worker profile is created with a client-generated UUID that is stable across retries; the server never invents identities for offline-created records.
- Idempotency-Key. Mutating requests carry a client key; the server guarantees that retransmitting the same key returns the original result and never duplicates state.
- Append-only evidence. Evidence records may be created and referenced but never mutated or deleted; corrections are new records that supersede.
- LWW with versioning. Mutable fields carry a monotonic version integer. A write applies only if its version exceeds the stored version, breaking ties deterministically by timestamp and UUID; this gives offline clients a well-defined merge semantics closer to CRDT behaviour [17].
- Resumable uploads. Binary evidence flows through the tus gateway so a dropped rural connection resumes from the exact byte offset rather than restarting [20].

Collectively these invariants let an assessor complete an entire evidence capture in a remote village without connectivity and synchronize later with no loss or duplication.

IV. Assessment Lifecycle: a Shared State Machine

All parties see one consistent lifecycle for an assessment, defined as a deterministic finite automaton (S, E, G, δ) where S is the set of states, E the event set, G the role guards and δ the transition function. The state set is

$$S = \{\text{Draft}, \text{Declared}, \text{Mapped}, \text{PlanReady}, \text{InProgress}, \text{Submitted}, \text{AiAnalyzed}, \text{UnderReview}, \text{Reassess}, \text{Reviewed}, \text{Approved}, \text{Rejected}\}, \quad (1)$$

mirroring the official Advise→Assess→Audit structure: Draft through PlanReady belong to the advise phase (QP mapping and assessment-plan generation); InProgress through Reviewed to the assess phase; and the audit phase is realized as post-certification sampling queries over Approved records. Figure 2 gives the full transition diagram. Guards are role-based: a worker may Declare and submit evidence; an assessor maps declarations to QPs, generates the plan, and issues the final decision; an authorized assessor may return an assessment to Reassess; an admin audits. All transitions are validated against the contract and recorded as events in the audit chain (Section VI-B).

V. Deterministic-First Analysis Engine

The analysis engine scores the evidence a candidate submits: a spoken or written narration of experience plus task evidence (photos/video of practical work). The governing design rule is that the decision path is deterministic: the system computes, from the QP contract alone, how well the narration and evidence match the performance criteria, and only then decides—by explicit thresholds—whether a learned model may add signal.

A. Bilingual keyword scoring

Each performance criterion pc carries a keyword set K_{pc} plus, for the regional packs, language-specific keyword sets $K_{pc}^{(hi)}$ and $K_{pc}^{(mr)}$ in Devanagari script, authored from the same QP source. A normalization function

$$N(r) = \text{strip}_{\text{punct}}\big(\text{fold}(\text{NFKC}(r))\big) \quad (2)$$

lowercases, Unicode-normalises, and removes punctuation and hyphen delimiters so that "two-way-switch" and "two way switch" collide. A criterion is deemed matched in language ℓ when any of its keywords appears in the narration token set

$$M_\ell(pc, r) \;=\; \mathbf{1}\Big[\, K_{pc}^{(\ell)} \cap \text{tokens}\big(N(r)\big) \neq \emptyset \,\Big], \quad (3)$$

where $\ell \in \{\text{en}, \text{hi}, \text{mr}\}$, $K_{pc}^{(\text{en})} = K_{pc}$, and the spoken-language keyword sets are active only when the narration script is Devanagari. English matching is always active, so a bilingual narration in Hindi with English trade terms scores against both planes without double counting. This is a deliberate lexical first pass: it is explainable, deterministic, and immune to model drift, and it does not require labelled training data, which RPL populations lack.

B. Coverage, confidence and routing

Per-language coverage and the overall confidence are

$$C_\ell(r) = \frac{1}{|\mathcal{P}_r|} \sum_{pc \in \mathcal{P}_r} M_\ell(pc, r), \qquad \text{conf}(r) = \max_\ell C_\ell(r), \quad (4)$$

with $\mathcal{P}_r$ the PC set of the candidate's mapped NOS. The router is explicit:

$$\rho(r) = \begin{cases} \text{ai_score}, & \text{conf}(r) \geq \tau_1 \\ \text{llm_augment}, & \tau_2 \leq \text{conf}(r) < \tau_1 \\ \text{human_assess}, & \text{conf}(r) < \tau_2 \end{cases} \quad (5)$$

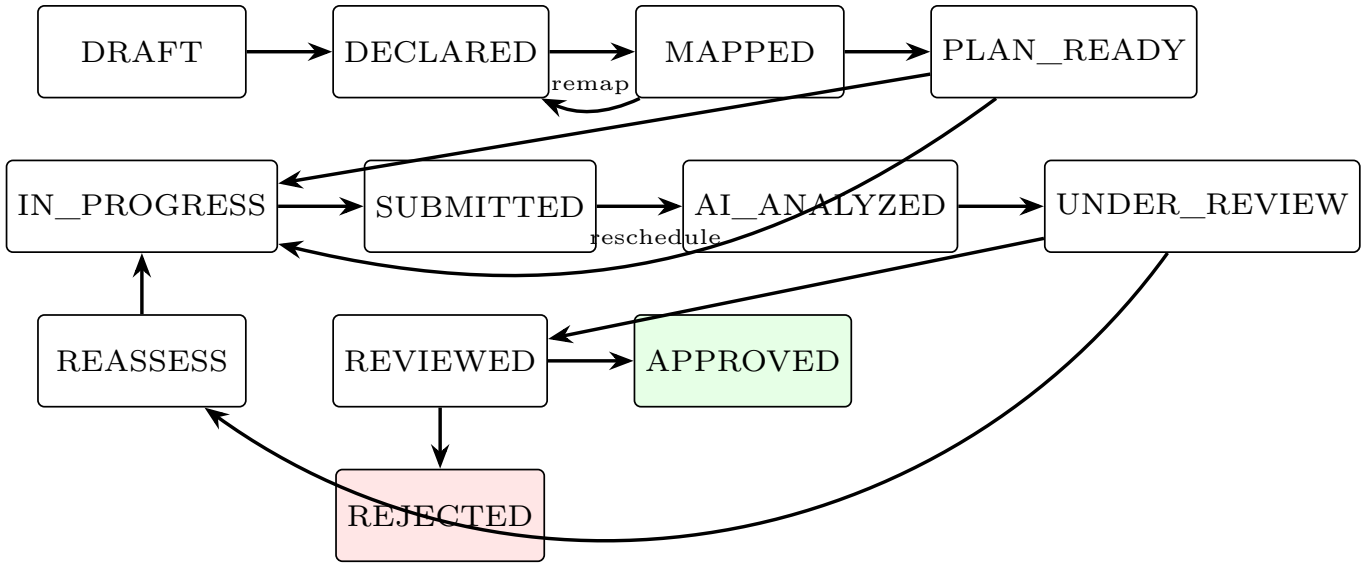


Fig. 2. Shared assessment state machine. DRAFT–PLAN_READY implement the NSQF advise phase, IN_PROGRESS–REVIEWED the assess phase; approbation and rejection are audit-phase guarded.

with $\tau_1 = 0.6$, $\tau_2 = 0.4$ (defaults, configurable per pack). When confidence is high the deterministic score stands; in the middle band a regional-language narration prompt asks the LLM to paraphrase the narration against the same PC rubric (never to decide); and low confidence always routes to a human assessor—unmatchable Devanagari narration is honestly passed to people rather than force-scored by a model that may not understand it. Adapters are behind capability flags: in deployment and in tests, AI_MODE=mock exercises the adapter interfaces deterministically.

C. Optional learned adapters

Three adapters upgrade signal without changing the deterministic core. (i) ASR: Whisper-style transcription converts regional-language audio narration to text which then enters the same deterministic scorer [14]. (ii) Multilingual retrieval: LaBSE-style embeddings rank candidate free-text experience descriptions against QP titles, proposing QPs for the advise phase and feeding Mapped [13]. (iii) Computer vision: YOLO-style models detect PPE and procedural artefacts in video frames sampled at rate ϕ; the adapter reports a detected safe-frame fraction $\kappa = |\text{safe frames}|/|\text{sampled frames}|$ as advisory evidence for the assessor, and any failure of a critical-safety PC still vetoes independently (Section VI) [15].

VI. Scoring, Certification and Auditability

A. Weighted scoring and critical veto

Each assessment plan selects blueprints (practical task specifications) $j = 1 \dots m$ with contract weights w_j normalised to sum to 100 and an assessor-assigned per-blueprint score $s_j \in [0, 1]$. The NOS-level practical score is

$$S_N = \frac{\sum_j w_j \, s_j}{\sum_j w_j}, \qquad \sum_j w_j = 100, \tag{6}$$

and the final score is a convex combination of theory and practical components, $S_{\text{tot}} = \alpha \, S_{\text{theory}} + (1 - \alpha) \, S_N$ with $\alpha \in [0.20, 0.30]$ per the QP's stated theoretical weight. Critical-safety performance criteria $\mathcal{P}_c$ (e.g. "isolate the circuit before opening the panel") act as a veto:

$$v = \prod_{pc \in \mathcal{P}_c} M_{\text{en}}(pc, r), \qquad S_N^{\text{final}} = v \cdot S_N. \tag{7}$$

The decision rule applies the NSQF RPL standard per NOS:

$$\text{decision} = \begin{cases} \text{Approved}, & S_N^{\text{final}} \geq \theta \;\; \forall N, \;\; \theta = 0.70 \\ \text{Bridge}, & \exists \, N : S_N^{\text{final}} < \theta, \; C_\ell(r) \geq \tau_2 \\ \text{Rejected}, & \text{otherwise} \end{cases} \tag{8}$$

Bridge cases emit a gap report with a deterministic bridge-training plan of

$$H_b = \lceil \, (1 - C_\ell(r)) \cdot H_{\max} \rceil \tag{9}$$

contact-hours over the specific unmet PCs, operationalising UNESCO's recommended bridge module and part qualification [1]. Every component of (6)– (9) is recomputable by a verifier from the stored contract and evidence.

B. SHA-256 evidence chain and public verification

Assessment artefacts are written append-only into an evidence chain. With H_0 derived from the contract schema version and seed, the chain update is

$$H_i = \text{SHA-256}(H_{i-1} \,\|\, \text{ts}_i \,\|\, \text{ev}_i \,\|\, \text{SHA-256}(B_i)) \,, \qquad (10)$$

where ts_i is a server timestamp, ev_i the event (e.g. submitted, ai_analyzed, approved), and B_i the payload bytes (submission, narration transcript, report, certificate blob). The chain binds ordering: changing any historical payload invalidates all subsequent hashes. This follows the admissibility logic of BSA 2023 §63, which treats hash-locked electronic records as secondary evidence only when their integrity can be demonstrated cryptographically [21], and provides the immutable audit trail that paper records cannot.

On approval the certification service renders a PDF certificate carrying a QR code pointing at a public verifier endpoint /verify/:certificateNo. The verifier page is served by the same backend, requires no authentication, recomputes the chain for the certificate, and displays the candidate's (optionally masked) name, the QP, the NSQF level, the approved date, and the per-NOS outcomes—so an employer, migration officer, or auditor can independently confirm that the issuer's evidence chain is intact. Domain hashing (SHA-256) and transport encryption follow standard practice (TLS 1.3) [22]; records at rest are encrypted with AES-256.

C. Client synchronization revisited

The LWW rule enforced by the API is, for each record key k,

$$\text{write}(k, d) \;\Rightarrow\; \max_{d' \in \mathcal{D}(k)} \left(v_{d'},\ \text{ts}_{d'},\ \text{uuid}_{d'}\right) \qquad (11)$$

compared lexicographically—the record with the greatest (version, timestamp, uuid) wins, making merges deterministic across offline replicas. Combined with the Idempotency-Key and append-only evidence rules of Section III, this guarantees that field capture followed by eventual sync never duplicates or loses state, independent of network partitions.

VII. Multilingual Assessment and Fairness

Regional-language assessment is not cosmetic: the NCVET RPL guidelines and UNESCO's India case study both call for assessment in regional and local languages, and candidate populations are disproportionately low-literacy [5], [1]. Three design choices support fairness. First, language-parallel rubrics: keyword sets for Hindi and Marathi are authored directly from the official QP text rather than machine-translated, so scoring semantics are anchored in the standard. Second, script-aware matching: the normalizer (2) keeps Devanagari glyphs intact and matching is performed in-language, so a Marathi narration is not silently compared against English keywords. Third, honest fallback: the router (5) refuses to fabricate a score when signal is genuinely absent—an unmatchable Devanagari narration goes to a human assessor rather than to a model that may hallucinate. Analytics over assessment results are computed only in the aggregate (per-pack pass rates, per-PC difficulty) and are never shown at the candidate level to assessors before a decision, preventing anchoring bias.

TABLE I
Trade packs shipped with the platform (contracts, not prose).

Pack	QP	NSQF	NOS	PC	BP	MCQ	EN	hi/mr
Electrician	PSS/Q6001	3	11	168	11	28	580	—
Welder	CSC/Q0204	3	5	80	10	24	402	—
Plumber	PSC/Q0104	4	8	159	12	24	509	167/60
Mason	CON/Q0103	3	8	162	12	24	498	147/147

PC = performance criteria, BP = blueprints, MCQs = theory bank, EN = English keywords, hi/mr = Hindi/Marathi keywords. Weights of blueprints sum to 100 per pack.

VIII. Evaluation

A. Implementation

The backend is implemented in Node.js/TypeScript (ESM) on Express 5 with Zod v4 contracts, Mongoose 9 on MongoDB, BullMQ 6 queues on Redis (with an inline driver for tests and single-host deploys), JWT role tokens, PDFKit + QR certificate rendering, and a tus media gateway. The bilingual analysis service, verifier page, and the Plumber/Mason trade packs are the hackathon additions over the base platform. The full test suite is Vitest + Supertest over in-memory MongoDB: 125 tests across 14 suites (auth, assessment lifecycle, analysis, bilingual scoring, evidence chain, sync/LWW, contract, tus resumable upload, trade-pack schema integrity, report/certificate and public verifier, review, admin, profile, health).

B. Trade-pack corpus

Four QPs are fully encoded as machine-readable contracts (Table I), parsed and validated at load time by the shared loader—no pack can be shipped that violates the schema (unresolved internal references, missing keywords, weight sums different from 100, fewer than 24 MCQs, or missing one of the six required task types are load-time failures). The corpus totals 32 NOS, 569 performance criteria, 45 blueprints, 100 MCQs, and 1,989 English keywords, of which the Plumber pack alone contributes 167 Hindi and 60 Marathi keyword sets—the first published bilingual RPL assessment contracts for these trades to our knowledge. Data was transcribed from the official CSDCI/PSSC facilitator guides and recorded in each pack's provenance field, keeping the mapping auditable. Strict linting (minimum 3 keywords per PC, per-PC uniqueness, resolved references, exact weight sums) runs in CI.

C. Functional verification

The suite exercises the full lifecycle from Draft through Approved/Rejected: role-guarded transitions (a worker cannot map a QP; an assessor decides); the bilingual scorer across English, Hindi and Marathi narrations with punctuation-insensitive matching; router behaviour (high-confidence score, mid-band LLM augment, low-confidence human route); the critical-safety veto (a missing "isolate the circuit" criterion fails the NOS regardless of other scores); append-only enforcement; LWW merge determinism; tus resume at a byte offset; the 70% rule; certificate issuance with QR and public /verify resolution; and 404 on forged certificate numbers. All 125 tests pass with AI_MODE=mock and the inline queue driver; learned adapters are contract-tested through their interfaces but not statically evaluated here.

D. Limitations

We stop short of (i) a production vision/ASR stack (adapters are interface-complete and mocked), (ii) field pilots—reported domain figures such as 15–22% inter-assessor variation or 94.2% vision mAP are literature values, not measurements of this platform—and (iii) the PWA client build, whose synchronization semantics are specified and enforced server-side. Turnaround and consistency targets are engineering targets to be validated in pilots.

IX. Conclusion and Future Work

We presented a deterministic-first, contract-driven RPL assessment platform that answers three structural weaknesses of deployed RPL: subjective manual scoring, regional-language exclusion, and offline evidence capture without verifiability. The central trade—trusting a machine-verifiable rubric computed from the official QP, with learned models relegated to explicit, gated advisory roles—makes the assessment explainable and auditable, which is exactly what certification bodies and courts require. The system ships four NSQF-aligned trade packs, a bilingual scoring engine with honest human routing, an offline-first sync contract, and a public certificate verifier over a SHA-256 evidence chain, validated by 125 automated tests over 37 API paths.

Future work proceeds along four lines: (1) field pilots with a partner SSC to measure inter-assessor consistency and turnaround against the engineering targets; (2) production vision/ASR adapters with calibrated uncertainty; (3) native part-qualification and bridge training flows as first-class states; and (4) publication of generated OpenAPI packs and bilingual rubrics back to the National Qualification Register ecosystem, so that any certifying body can operate the engine against its own QPs.

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